indicates that NeuroCodec's perceptual advantage is not solely attributable to its wider output bandwidth and is consistent with benefits from the proposed extraction architecture. The substantially higher score of the original NeuroCodec output further demonstrates the perceptual benefit of wider-band speech reconstruction. The mixture obtained a mean score of 22.41 (CI: 20.24-24.80). The NeuroCodec-M3ANet difference was significant under a paired permutation test ($p = 3.0 \times 10^{-5}$).

D. OnlineNeuroCodec Evaluation

In addition to the offline evaluation, we assess OnlineNeuroCodec against our reimplementation of online NeuroHeed [15].[1] OnlineNeuroCodec and the reimplemented NeuroHeed are both evaluated under an identical streaming configuration with a window length of $w = 2.0$ s and a hop size of $w_c = 0.5$ s, with all benchmarks measured on a single GTX 1080 Ti GPU. Computational efficiency is evaluated using the Real-Time Factor (RTF $= T_{\text{proc}}/w_c$), where T_{proc} is the processing time. We note that NeuroHeed [15] defines RTF using the reciprocal convention, w_c/T_{proc}.

As shown in Table III, OnlineNeuroCodec attains substantially higher extraction accuracy. Relative to NeuroHeed [15], STOI improves from 0.612 to 0.803, ESTOI from 0.5716 to 0.7105, and PESQ from 1.290 to 1.662.

TABLE III
ONLINE EVALUATION OF ONLINENEUROCODEC AGAINST NEUROHEED [15] ON THE KUL DATASET.

Model	STOI↑	ESTOI↑	PESQ↑	RTF↓
NeuroHeed [15]	0.588	0.497	1.037	0.27
NeuroHeed [15] *(with autoregressive Speaker Encoder)*	0.612	0.572	1.290	0.41
OnlineNeuroCodec	0.742	0.686	1.501	**0.13**
OnlineNeuroCodec *(with Auditory Attractor)*	**0.803**	**0.711**	**1.662**	0.18

NeuroHeed results are from our reimplementation, as the authors release code for the offline model only.

Enabling the auditory attractor improves OnlineNeuroCodec from 0.742 to 0.803 STOI and from 1.501 to 1.662 PESQ. Since $\mathbf{Z}_{\text{past}}$ affects the model only in mean-pooled form (Eq. 6), this improvement is consistent with more stable tracking of the attended speaker across hops. A comparable trend is observed for NeuroHeed, whose autoregressive speaker encoder raises STOI from 0.588 to 0.612, consistent with the self-enrollment behaviour reported in their work [15].

OnlineNeuroCodec without the attractor achieves the lowest RTF (0.13). Enabling the auditory attractor increases RTF modestly from 0.13 to 0.18. NeuroHeed is the most expensive of the four configurations (0.27-0.41). OnlineNeuroCodec is therefore approximately 2.1× faster than NeuroHeed without the attractor mechanism (0.13 against 0.27) and 2.3× faster with it (0.18 against 0.41), so its accuracy advantage is obtained at approximately half the measured RTF, despite operating on a generative rather than a masking objective.

[1]The public NeuroHeed repository provides only the offline implementation. CA-NeuroSpex [20] is omitted because no public implementation is available for reproduction.

Table IV reports the effect of the two streaming parameters. Lengthening the buffered context w_b improves PESQ, since a longer causal history is available for cross-modal alignment, but raises the RTF because each hop must reprocess a longer window. The gain in PESQ saturates beyond $w_b = 1.5$ s while the RTF continues to grow. Enlarging the hop w_c lowers the RTF, as fewer forward passes are required per second of emitted audio, but increases algorithmic latency because the system must wait for a full hop of new audio before it can respond. Hence, the configuration $w_b = 1.5$ s and $w_c = 0.5$ s is adopted throughout, as it attains near-maximal PESQ while maintaining a low RTF.

TABLE IV
EFFECT OF THE BUFFERED CONTEXT SIZE w_b AND THE HOP SIZE w_c ON ONLINENEUROCODEC.

	w (s)	PESQ ↑	RTF ↓
(a) Buffered context w_b ($w_c = 0.5$ s)			
$w_b = 0.5$ s	1.0	1.412	**0.09**
$w_b = 1.0$ s	1.5	1.558	0.13
$\mathbf{w_b = 1.5}$ **s**	**2.0**	**1.662**	**0.18**
$w_b = 2.5$ s	3.0	1.681	0.27
$w_b = 3.5$ s	4.0	1.689	0.36
(b) Hop w_c ($w_b = 1.5$ s)			
$w_c = 0.10$ s	1.6	1.674	0.71
$w_c = 0.25$ s	1.75	1.669	0.31
$\mathbf{w_c = 0.50}$ **s**	**2.0**	**1.662**	**0.18**
$w_c = 1.00$ s	2.5	1.594	**0.11**

The model was evaluated on the KUL dataset with the auditory attractor enabled. The total window is $w = w_b + w_c$. Bold text denotes the operating point used in Table III.

VI. CONCLUSION

In this paper we introduce NeuroCodec, a generative EEG-guided target speaker extraction framework that predicts latent representations of a frozen high-fidelity audio codec instead of estimating time-frequency masks. By combining graph-based EEG modeling, cross-modal attention, and Mamba-based temporal modeling, NeuroCodec achieves effective EEG-audio fusion in a compact latent space. Experiments on two datasets demonstrate improved speech intelligibility and perceptual quality over existing approaches, as measured by both objective metrics and a subjective listening study. NeuroCodec also enables 44.1kHz reconstruction with spectral content extending to approximately 20 kHz with minimal speaker interference, unlike prior work with limited output bandwidth and speaker leakage. We extend the model to a causal streaming version, OnlineNeuroCodec, which enables streaming TSE using a causal auditory attractor derived from previous target estimates. This model achieves vastly superior computational speed while retaining competitive extraction quality. These results demonstrate the potential of generative latent modeling for neuro-steered speech extraction and future assistive hearing systems. However, OnlineNeuroCodec, like most other contemporary models, remains far from deployment on embedded hearing-aid hardware. Future work will explore autoregressive codec-token prediction for fully causal generation with reduced buffering requirements, potentially lowering algorithmic latency, alongside model compression and low-power implementation.